

COVID Public Health Dashboard

Project Overview

This project analyzes COVID-19 trends using open public health data from Our World in Data (OWID). The objective was to compare COVID-19 cases, deaths, and vaccination progress across four countries: Canada, United States, India, and the United Kingdom. Python was used for data cleaning and preprocessing, while Tableau was used to build an interactive dashboard for visualization.

Project Goal

Compare COVID-19 trends between Canada and three other countries by analyzing new cases, total cases, deaths, and vaccination progress using Python and Tableau.

Tools Used

Python, Pandas, Matplotlib, Tableau Public, Brackets, Microsoft Excel.

Dataset

Our World in Data (OWID) COVID-19 Dataset. Data source: Our World in Data (CC BY).

Data Preparation

Loaded the dataset with Pandas, filtered Canada, United States, India, and United Kingdom, selected date, country, total_cases, new_cases, total_deaths, and people_vaccinated columns, converted dates, handled missing values, and exported cleaned_covid_data.csv for Tableau.

Dashboard Features

- New COVID-19 Cases Over Time
- Total Cases by Country
- Vaccination Trend
- Total Deaths by Country
- Interactive Country Filter

Key Insights

1. The United States recorded the highest total COVID-19 cases among the selected countries.
2. Canada and the United Kingdom showed strong vaccination progress over time.
3. All countries experienced multiple COVID-19 waves with different peak periods.
4. Differences in cases and deaths reflect variations in population size and public health responses.
5. Interactive dashboards help compare trends and support public health decision-making.

Conclusion

This project demonstrates a complete data analytics workflow, from cleaning raw public health data in Python to building an interactive Tableau dashboard that communicates meaningful insights.

Files Submitted

COVID_Analysis.py
cleaned_covid_data.csv
Tableau Dashboard (.twb/.twbx)
Dashboard Screenshot

COVID_Project_Summary.pdf

Resume Line

COVID Public Health Dashboard: Analyzed COVID-19 public health data using Python (Pandas, Matplotlib) and Tableau to compare case, death, and vaccination trends across Canada, the United States, India, and the United Kingdom by building an interactive analytical dashboard.

Data Source Credit

Data source: Our World in Data (OWID), licensed under Creative Commons Attribution (CC BY).